

“WHY SHOULD I TRUST YOU?” EXPLAINING THE PREDICTIONS OF ANY CLASSIFIER!

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PROBLEM: TRUST IS THE BOTTLENECK

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Focus of the paper:

Trust in NLP classifiers. Machine learning, especially neural models, has become central to NLP. But models are often black boxes. Understanding the reasons behind predictions is important in assessing trust in a model.

→ *The paper argues that if users do not trust a model or a prediction, they will not use it.*

Why Should I Trust You?

IMPLEMENTED TASKS AND MODELS

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→ The paper proposed to demonstrate the study on different NLP tasks like:-

- Politeness detection — for sentences.
- Multi-class content classification —for 20 Newsgroups dataset(Christianity vs. Atheism).
- Sentiment analysis — for sentences from movie reviews.

→Implemented Models: LSTMs, SVMs and Random forests trained on bag-of-words or on word embeddings.

→*In the Newsgroups dataset Model SVM with RBF(Radial Basis Function) kernel achieves 94% accuracy.*

ACCURACY ALONE IS NOT ENOUGH

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The paper explains accuracy can mislead due to:

- *Data leakage*: signal leaks into training/validation data.
- *Dataset shift*: training data is different than test data.
- *Metric mismatch*: computable metrics such as accuracy may not match user engagement, retention.

→ *Need trustful, interpretable explanations.*

→ *High accuracy does not guarantee trustworthy behavior of the model prediction.*

IMPLEMENTED EXPLAINABLE(XAI) MODELS

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The paper implements *LIME* and *SP-LIME* for local and global explanations.

→ *LIME* (Local Interpretable Model-agnostic Explanations): Explain Individual Predictions

- Explains any classifier locally.
- Get black-box predictions.
- Fit a simple, sparse interpretable model.
- Use human-understandable representation, e.g. word presence.

→ *LIME explains why a single prediction was made.*

IMPLEMENTED EXPLAINABLE MODELS (CONTINUED...)

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→ *SP-LIME*(Submodular Pick for LIME): Explain the global predictions.

- Selects a representative set of instances for explanation.
- Builds importance matrix of interpretable components.
- Maximizes coverage of important features.
- Gives a global, non-redundant view of the model.

→ The paper *was demonstrated LIME on multiple NLP tasks and classifiers, showing it is model agnostic and useful for trustworthy analysis.*

MOTIVATION FROM THIS PAPER (REPRODUCTION)

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Telecommunication customers churn prediction using hybrid neural network and Interpretable ML(XAI) methods.

Motivations:-

- Black box behavior of the Neural networks.
- Accuracy only is not a measurable factor.
- How/why telecom decision makers believe the hybrid neural network model prediction only?
- How LIME explains our model locally after training(post-hoc) in telecom dataset?
- How LIME agrees with other XAI methods like: SHAP and PDP in trustworthy explanation?



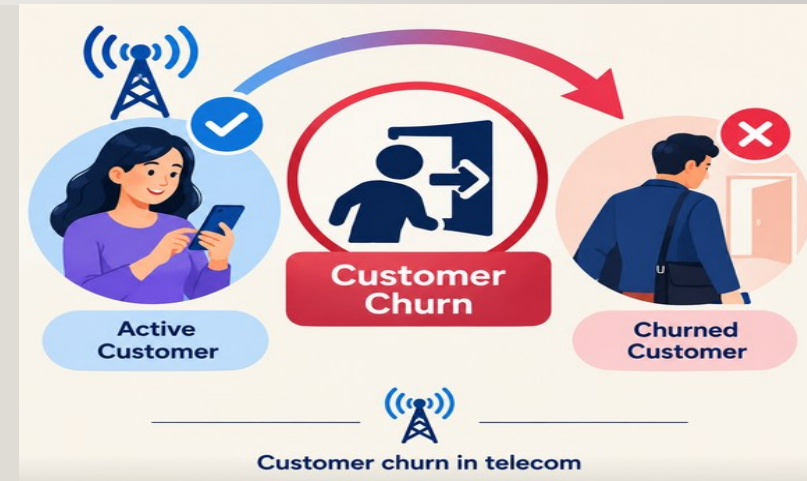
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PROPOSED HYBRID NEURAL NETWORK

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Hybrid model: CNN-BiLSTM-HybridAttention model.

- Performance comparison metrics:
→ (Accuracy, Precision, Recall, F1-Score, AUC).
- The hybrid model shows best results:
→ (Accuracy 82.04%, F1-Score 67.52% and AUC 0.86).



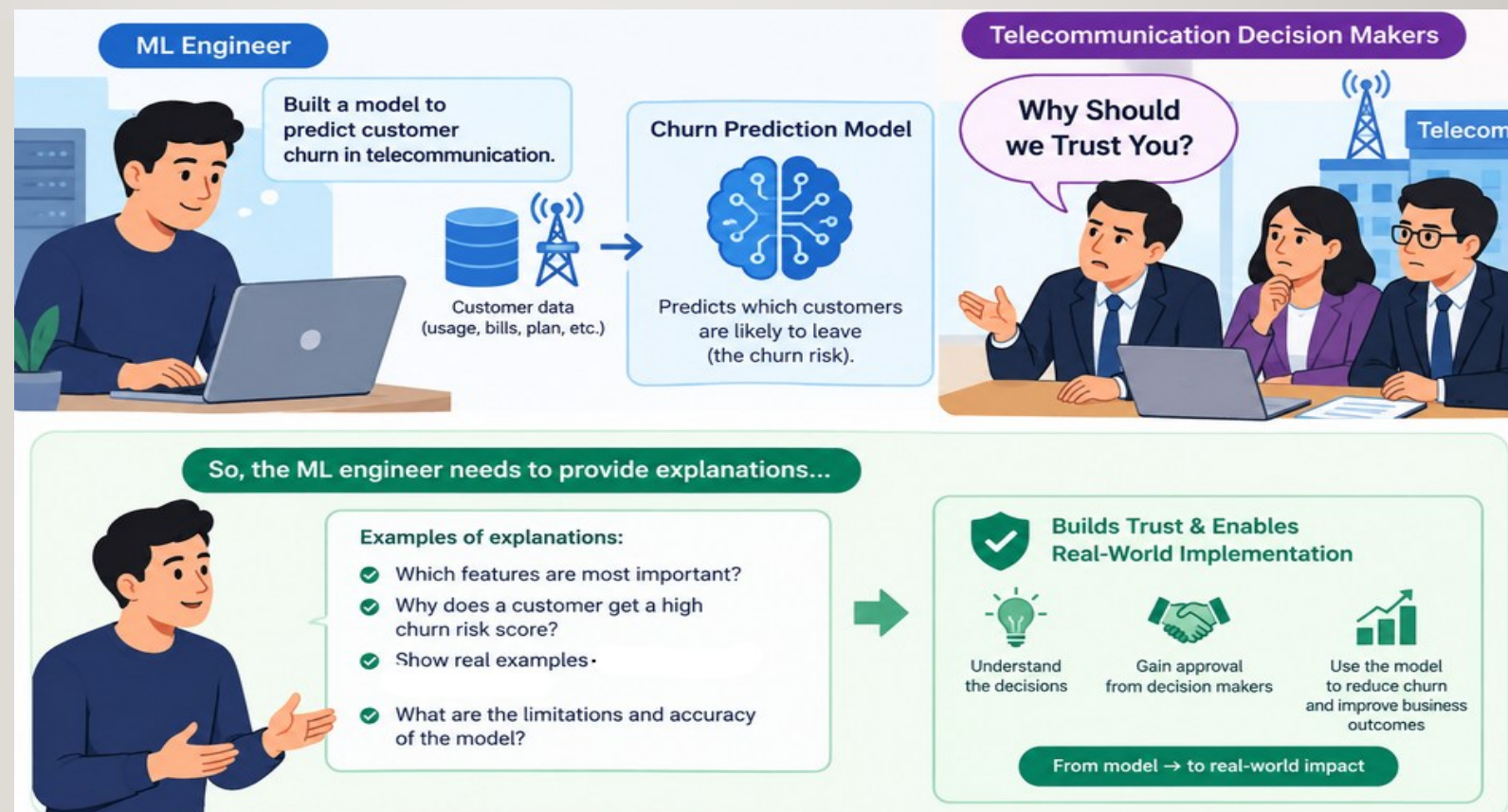
Model	Accuracy	Precision	Recall	F1-Score	AUC
CNN	77.36%	54.96%	80.16%	65.21%	0.86
BiLSTM	62.60%	41.07%	94.91%	57.33%	0.85
CNN-BiLSTM-HybridAttention	82.04%	64.78%	70.51%	67.52%	0.86

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WHY WE SHOULD WE TRUST YOU?

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Explain for us?



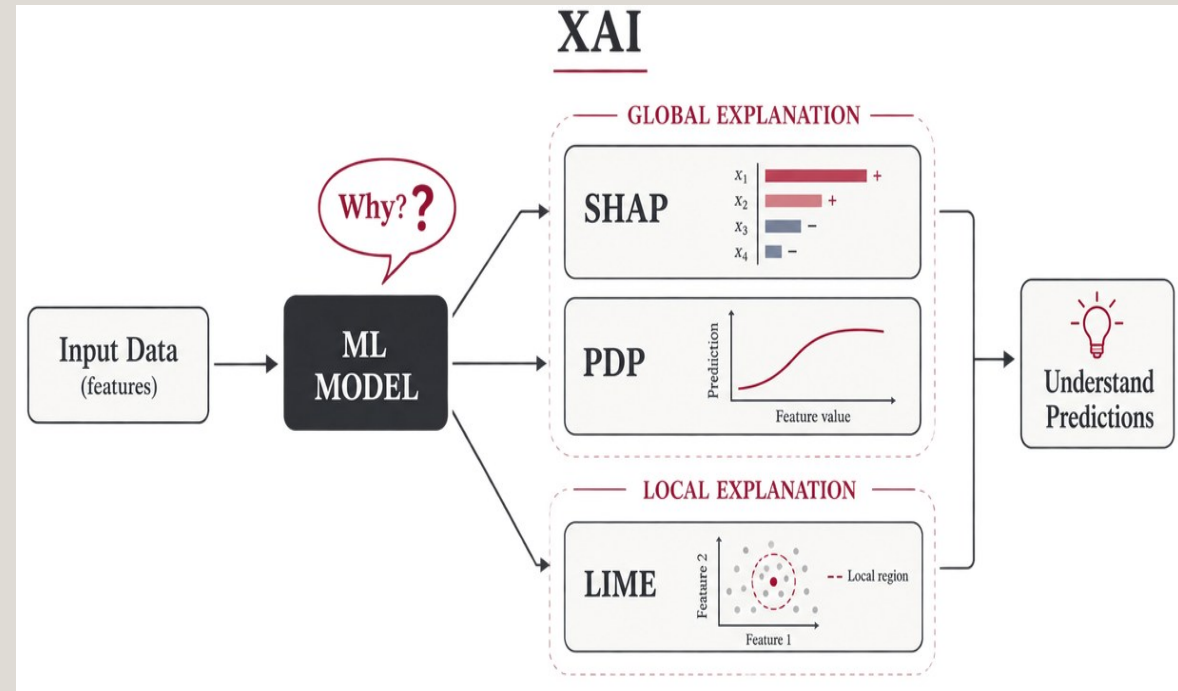
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EXPLANATION USING EXPLAINABLE(XAI) METHODS. AFTER TRAINING (POST-HOC EXPLAINABILITY)

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Triangulated explainability using:

- SHAP, Partial Dependence Plot (PDP), LIME.
- Global Explainability:
 - SHAP summary plot and PDP.
- Local Explainability:
 - SHAP waterfall plot, LIME Bar chart plot.



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GLOBAL EXPLAINABILITY USING SHAP (SHAPLEY ADDITIVE EXPLANATIONS)

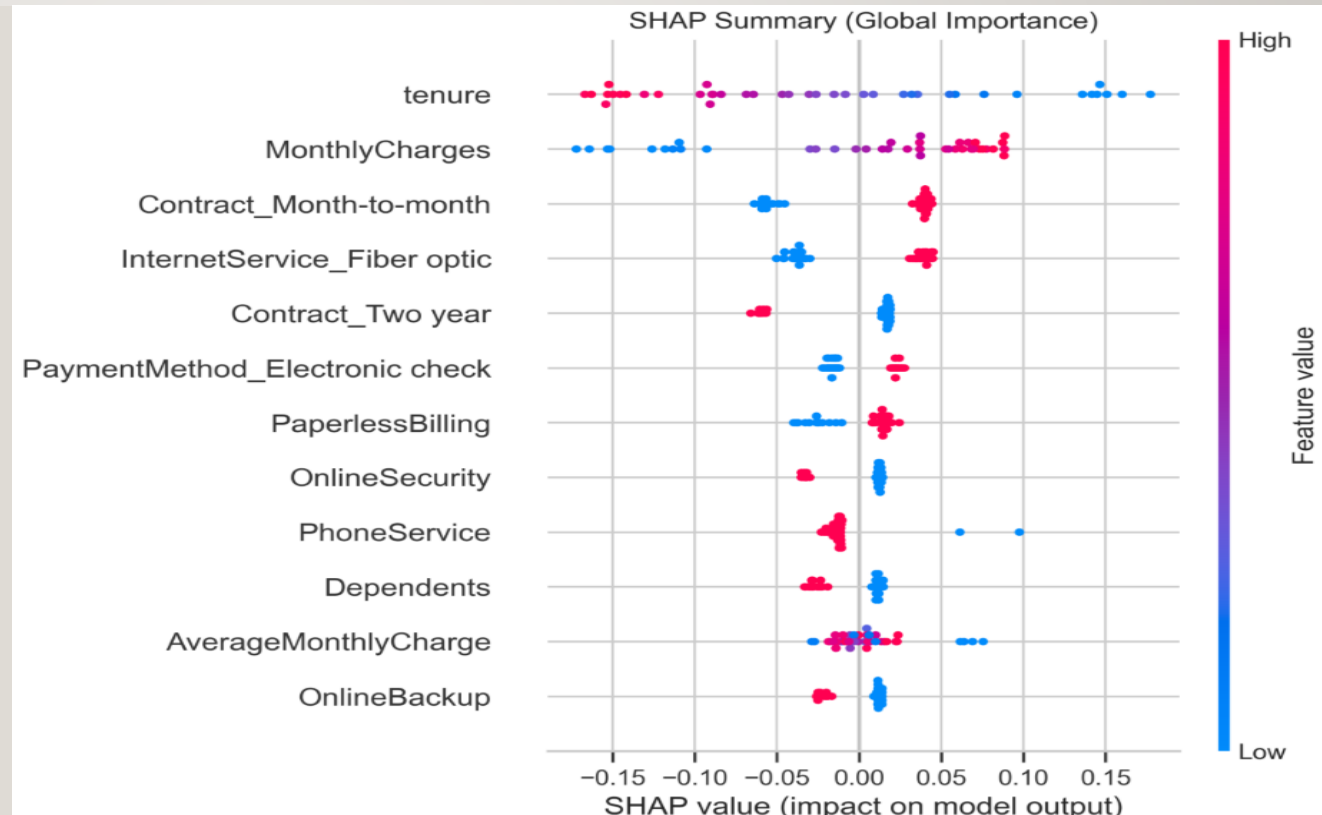
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SHAP Summary plot

→ Explains the entire model's behavior across all customers.

- Red flagged colors:
→ Churn (Exit). High.

- Blue flagged colors:
→ Not-churn (Stay). Low.

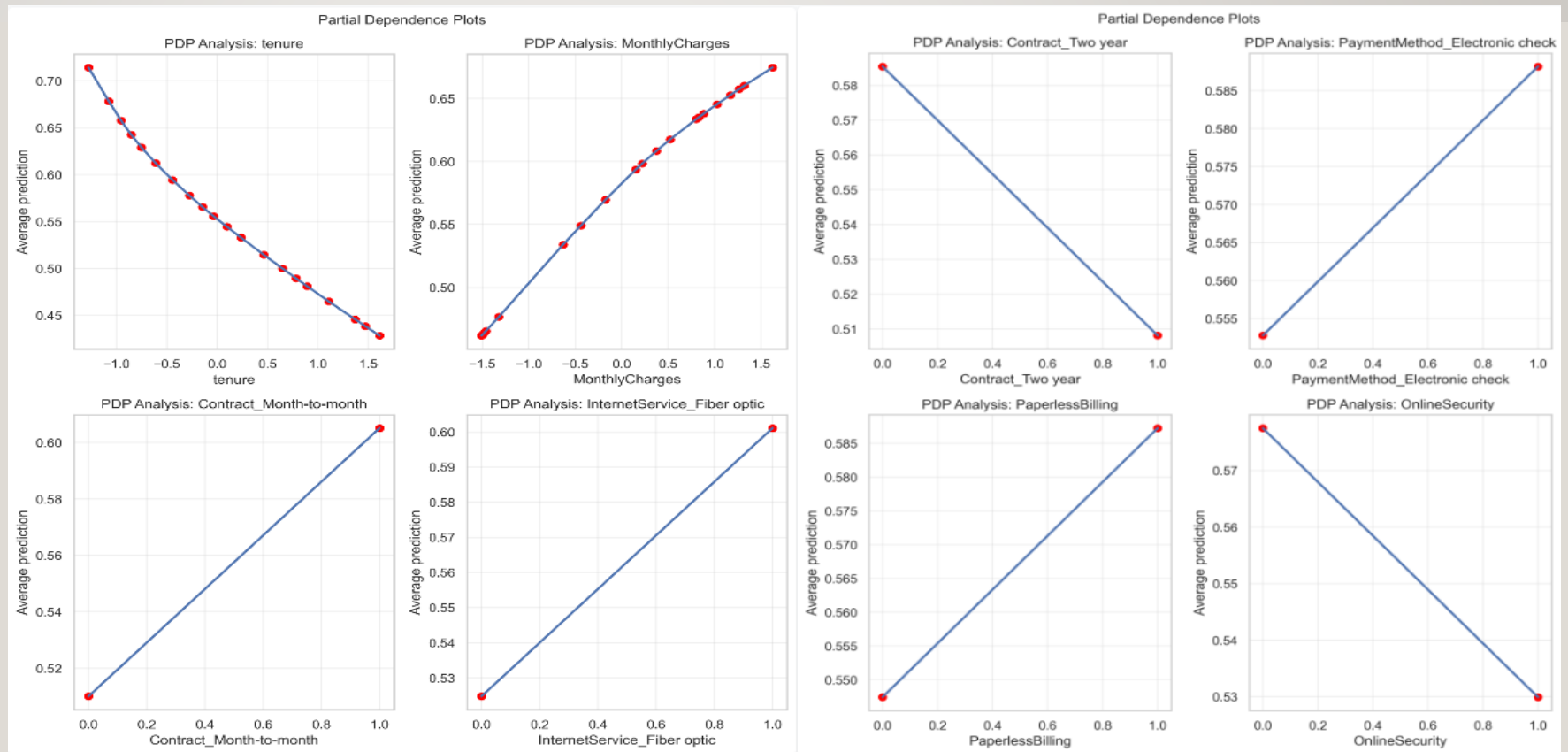


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GLOBAL EXPLAINABILITY USING PDP (PARTIAL DEPENDENCE PLOT)

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- Tenure (in months).
- Monthly Charges.
- Month to month contract.
- Fiber optic Internet service.
- Two years contract.
- E-check pmt. method.
- Paper-less billing.
- Online security.



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LOCAL EXPLAINABILITY

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Randomly selected customer:

→ Id: 5219-YIPTK.

→ Churned: Yes.

♦ Selected customer original file for local explainability ♦	
Feature	Value
customerID	5219-YIPTK
gender	Female
SeniorCitizen	0
Partner	Yes
Dependents	No
tenure	33
PhoneService	Yes
MultipleLines	Yes
InternetService	Fiber optic
OnlineSecurity	No
OnlineBackup	Yes
DeviceProtection	Yes
TechSupport	No
StreamingTV	Yes
StreamingMovies	Yes
Contract	Month-to-month
PaperlessBilling	Yes
PaymentMethod	Electronic check
MonthlyCharges	104.4
TotalCharges	3409.6
Churn	Yes

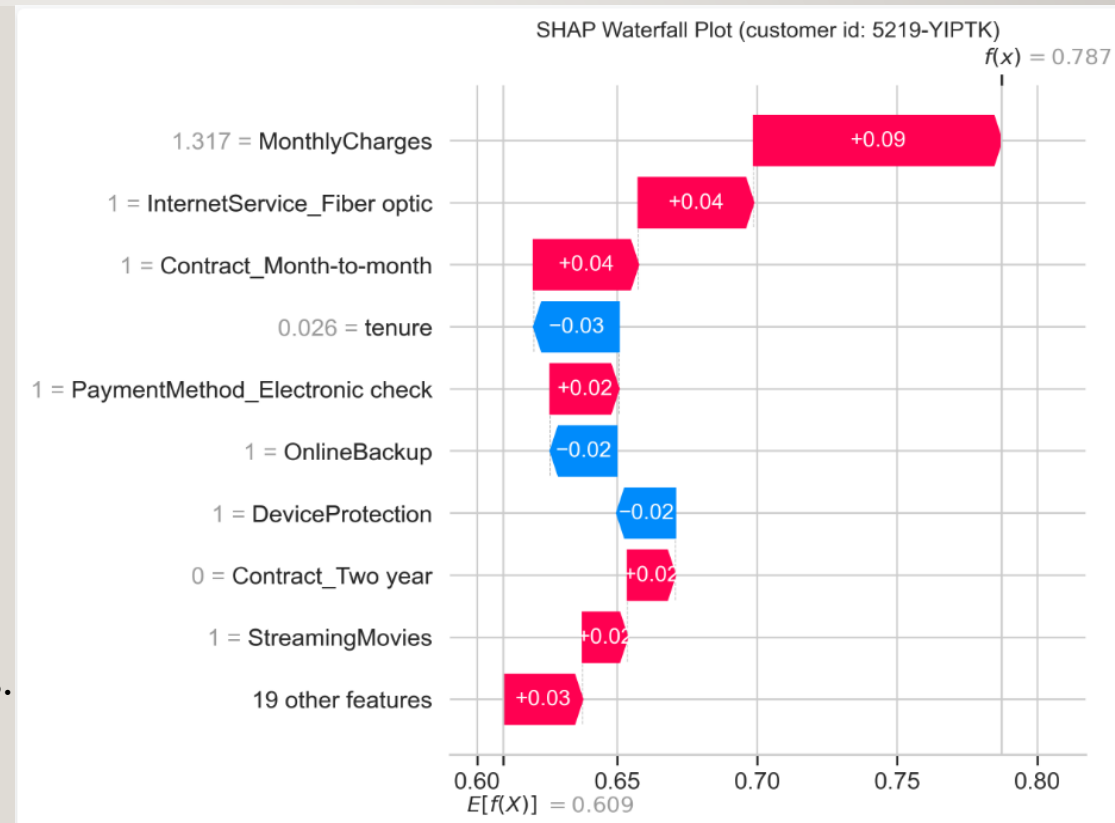
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LOCAL EXPLAINABILITY USING SHAP

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SHAP Waterfall plot

- Red flagged features (+ve):
→ **Churn (Exit).**
- Blue flagged features (-ve):
→ **Not-churn (Stay).**
- Bar-length — indicates feature importance.
- Expected Value (Base value), $E[f(X)] = 0.609$:
the average prediction across all customers.
→ Final prediction = Base value + feature contributions.
 $\approx 0.787 > 0.5 \rightarrow$ **Churned**



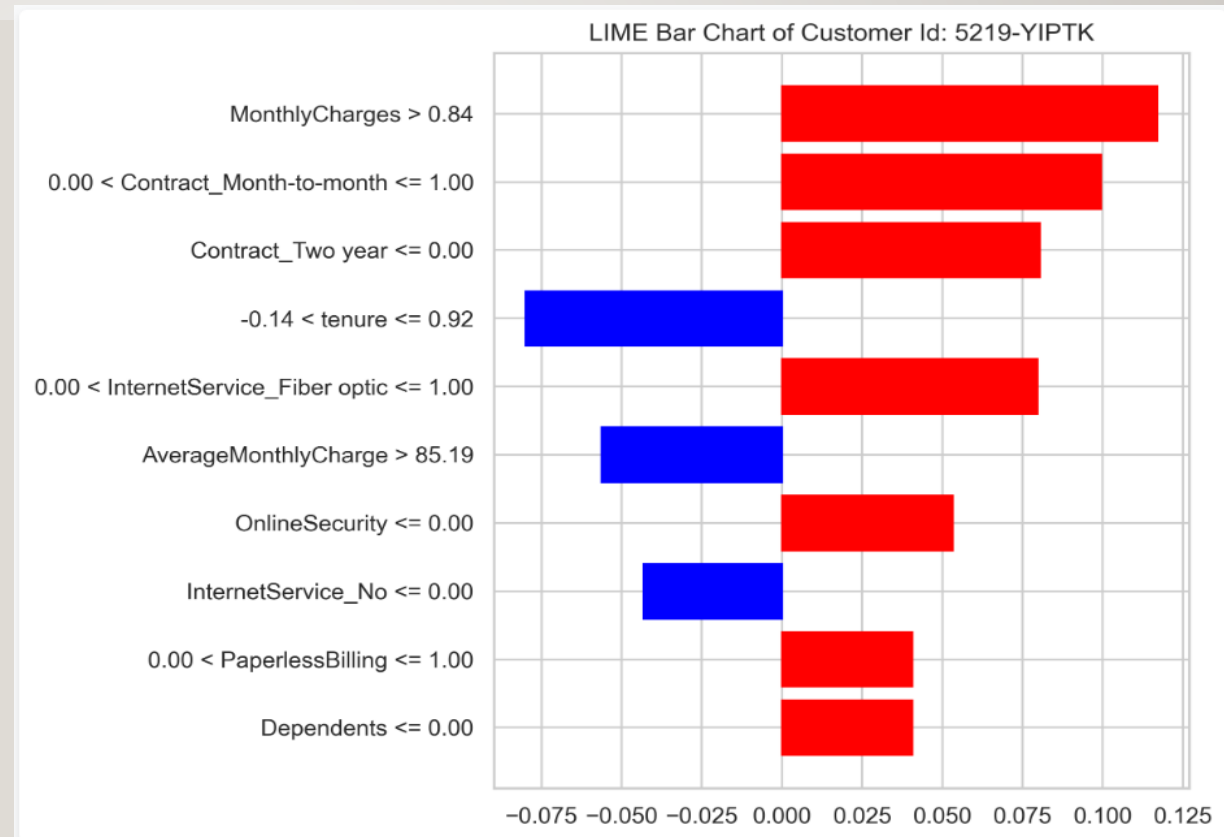
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LOCAL EXPLAINABILITY USING LIME (LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS)

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LIME Bar chart plot

- Red flagged features (+ve):
→ Churn (Exit).
- Blue flagged features (-ve):
→ Not-churn (Stay).
- Bar-length — indicates feature importance.
- Works under *IF* and *Then* Rule method.



- Explainability is done through triangulated XAI methods: SHAP, PDP, and LIME.
- Global interpretation identifies tenure, MonthlyCharges, and Contract_Month-to-month as top churn drivers.
- SHAP and LIME Local interpretation explains customer 5219-YIPTK is predicted to churn due to:
High MonthlyCharges, fiber optic Internet service, and month-to-month contract.
- *MonthlyCharges, Contract_Month-to-month, and InternetService_Fiber optic* are the most churn drivers across global and local explanations in this telecom dataset.
- *Tenure of months* is a major global driver and acts locally as a churn protective factor for selected customer.

- *SHAP* gives fair, consistent global and local explanation.
- *LIME* gives a quick, local, approximate, model-agnostic explanation.
- *LIME* agrees with *SHAP* on the key factors for the selected customer.
- Using *SHAP*, *PDP*, and *LIME* together makes the hybrid model's predictions more trustworthy and actionable.
- *Explanations*: answer “Why trust you?”, build trust and decision-maker approval, and turn churn prediction into real churn reduction and better business results.

LIMITATIONS

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Small Kaggle Telecom Dataset

 **7,043 subscribers**

Customer ID	Tenure (months)	Monthly Charges	Contract	Churn
1001	12	29.85	Month-to-month	No
1002	3	56.95	Month-to-month	Yes
1003	24	73.50	Two year	No
1004	8	59.00	Month-to-month	Yes
1005	47	23.75	One year	No

 **Open / public / limited scale**

Model trained on
small public data



Needs validation/
training on large
real-world data

Real-World Telecom Company Dataset

 **Millions of subscribers**



-  CRM (customer profiles)
-  Billing (invoices & payments)
-  Network usage (data & voice)
-  Call / data records
-  Recharge (top-ups)
-  Complaints (support tickets)
-  Service plans (current & historical)

 **Large / complex / continuously generated**



Limitation: results may not generalize to large, real-world subscription-plan environments.

Why Should I Trust You?

THE END!

